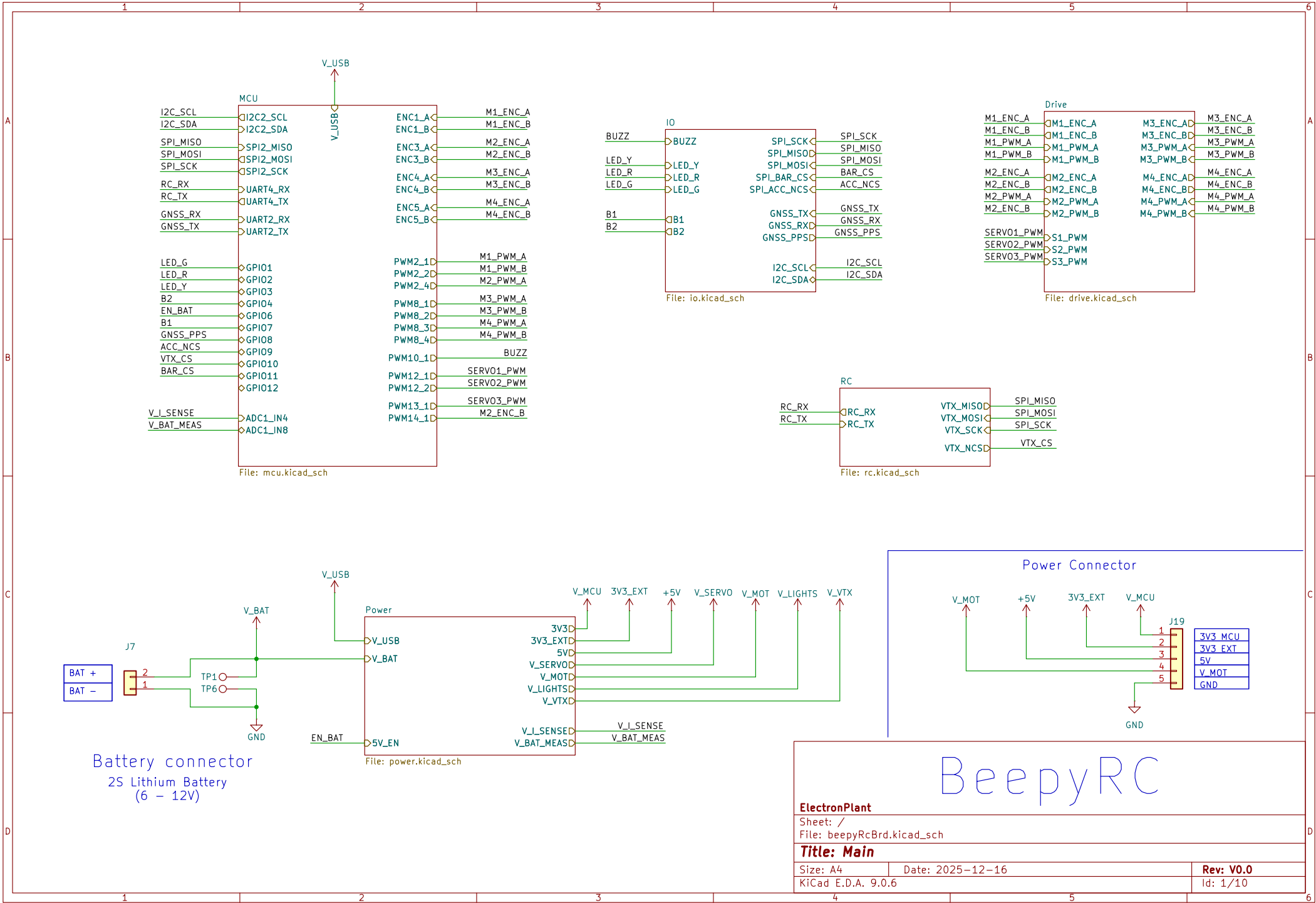
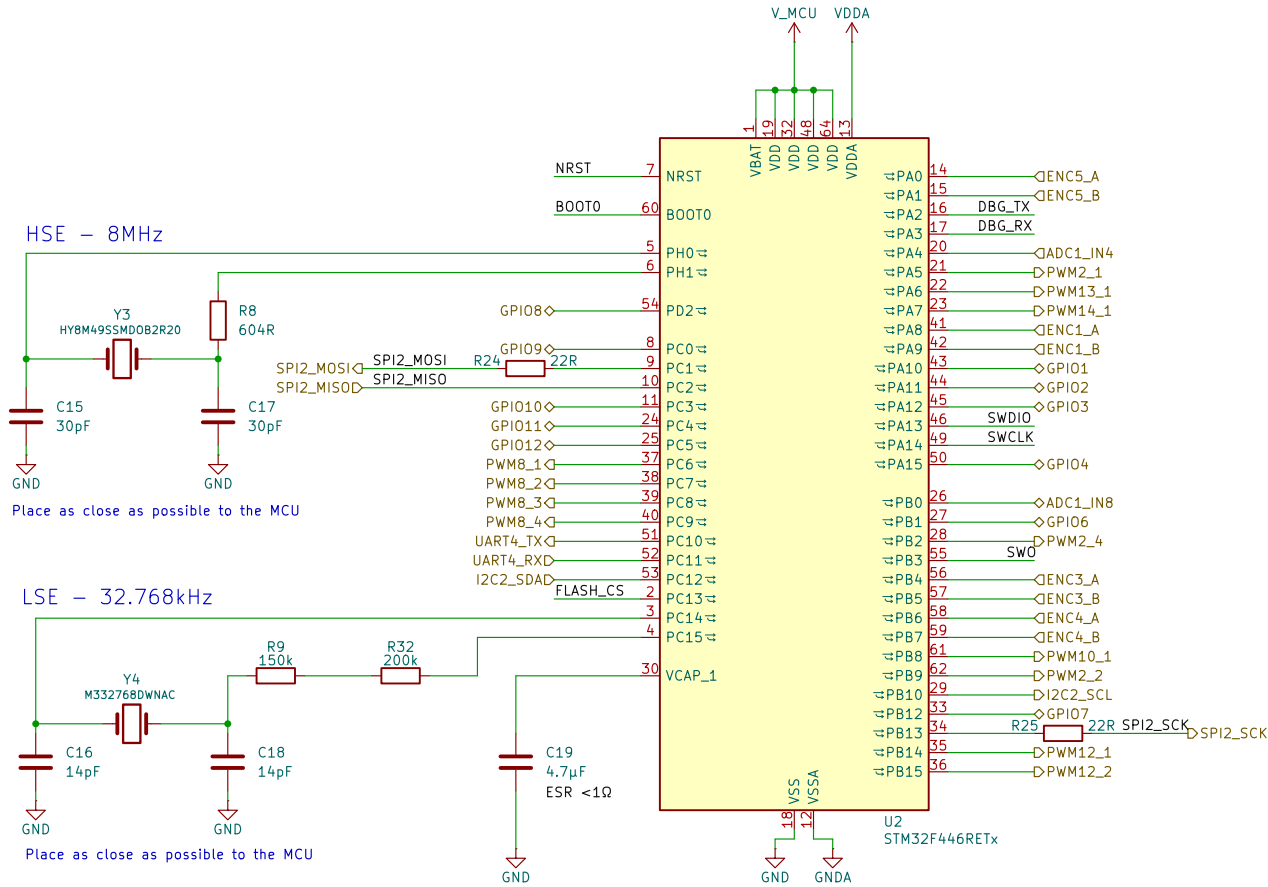




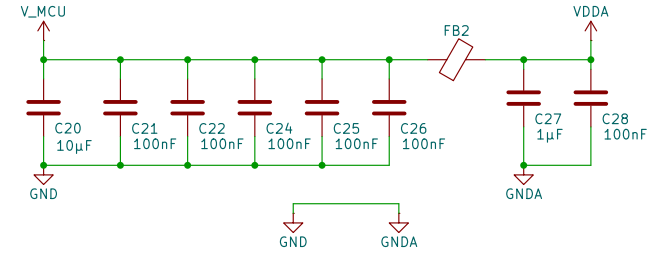
Battery connector
2S Lithium Battery
(6 – 12V)

BeepyRC

MCU



Power, Grounding & Decoupling

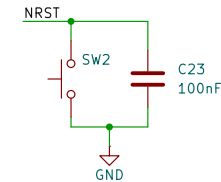


Decoupling strategy

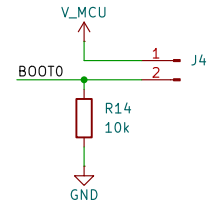
(Based on page 9 of AN4488)

- 10μF ceramic cap for the package.
- 100nF ceramic between pins 18-19, 31-32, 47-48 and 63-64.
- Vbat on pin 1 is connected to VDD with a 100nF decoupling cap.
- VDDA is connected to VDD through a ferrite bead.
- VDDA is decoupled with a 1μF and a 100nF cap between pins 12-13.
- GND is directly connected to the GND ground plane.

Reset

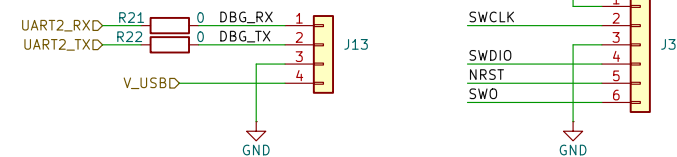


Boot

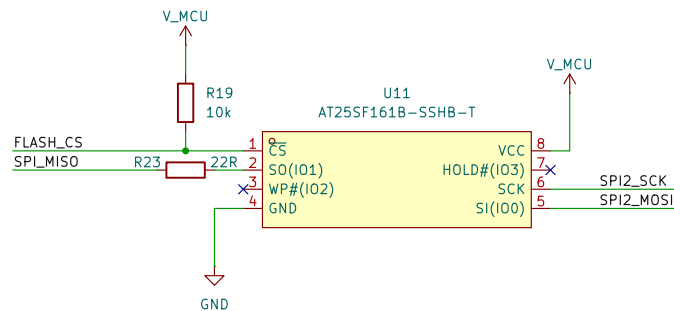


Debug

Debug Serial is shared



Flash (Config + Black Box)



BeePyRC

ElectronPlant

Sheet: /MCU/
File: mcu.kicad_sch

Title: MCU

Size: A4 Date: 2025-12-28

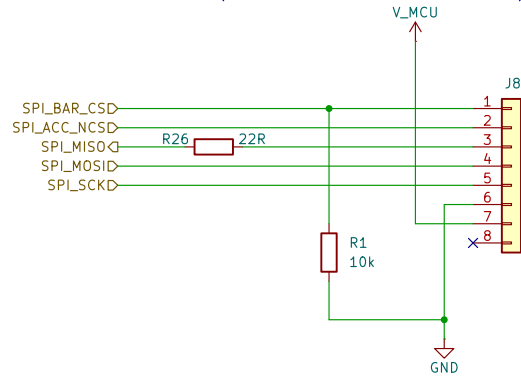
KiCad E.D.A. 9.0.6

Rev: V0.0

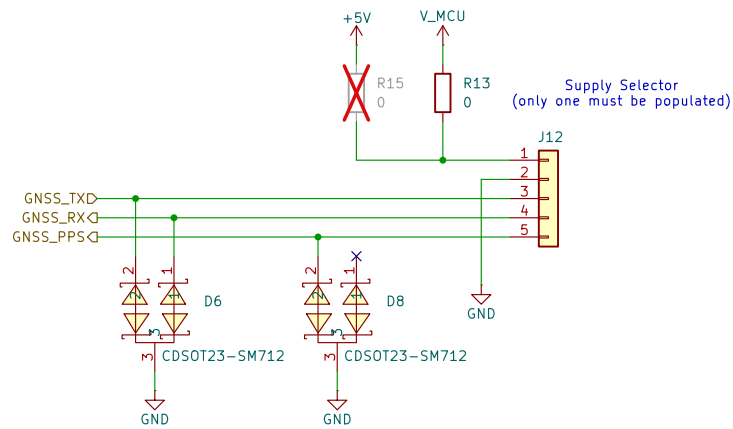
Id: 2/10

Sensors

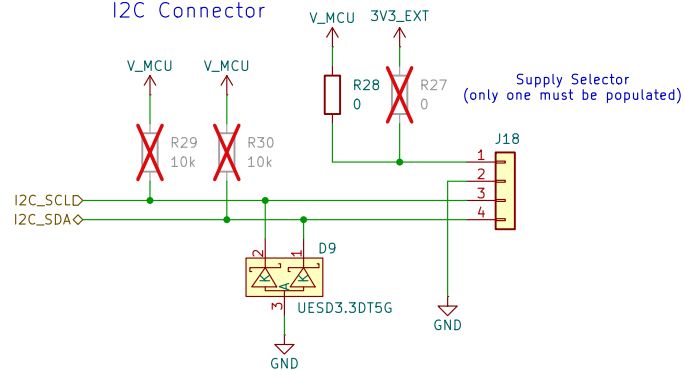
GY-91 (9-axis MPU9250 + BMP280) Module



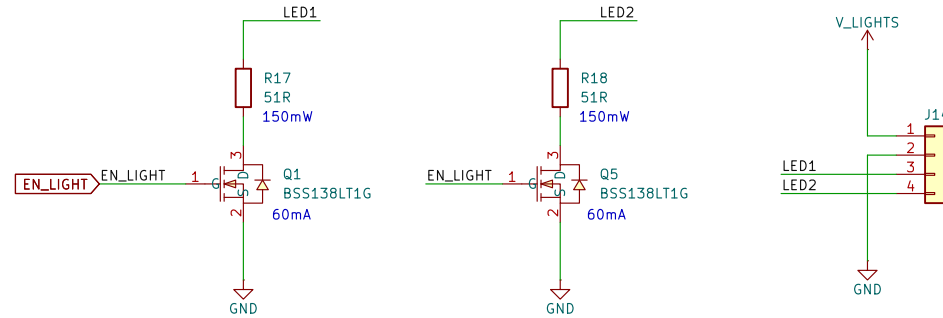
GNSS Module (L76K/VG7669T160N0MA)



I2C Connector

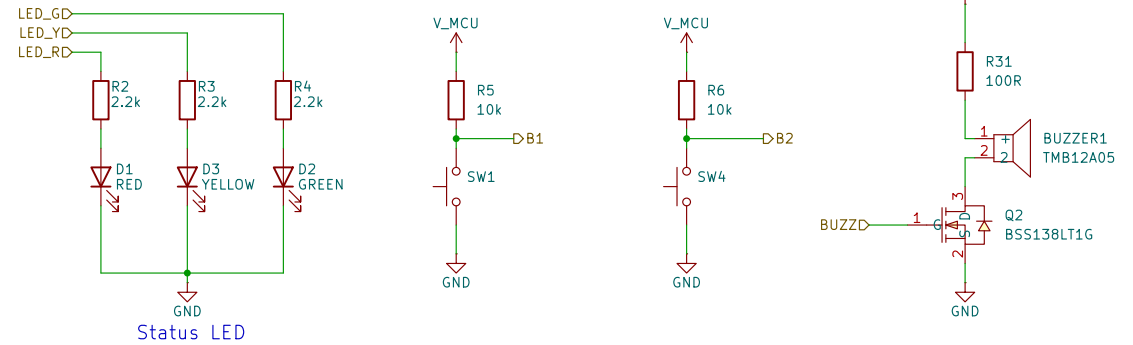


Lights



Current limiting resistor sizing:
- LED forward voltage: ~3V (2.8 - 3.4)
- Max forward current: 60mA

User Interface



BeepyRC

ElectronPlant

Sheet: /IO/

File: io.kicad_sch

Title: IO

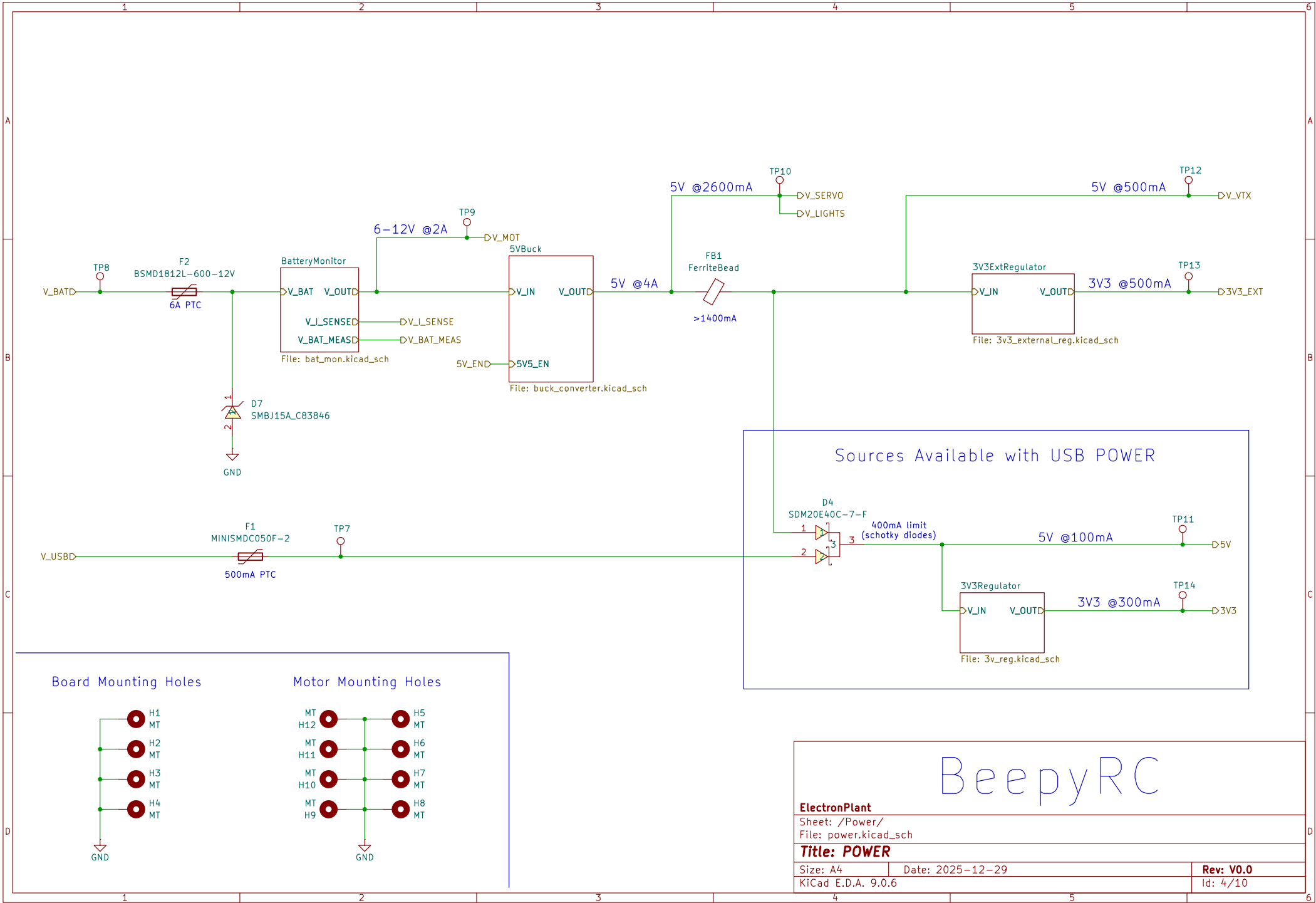
Size: A4

Date: 2025-12-29

KiCad E.D.A. 9.0.6

Rev: V0.0

Id: 3/10



BeepyRC

ElectronPlant

Sheet: /Power/
File: power.kicad_sch

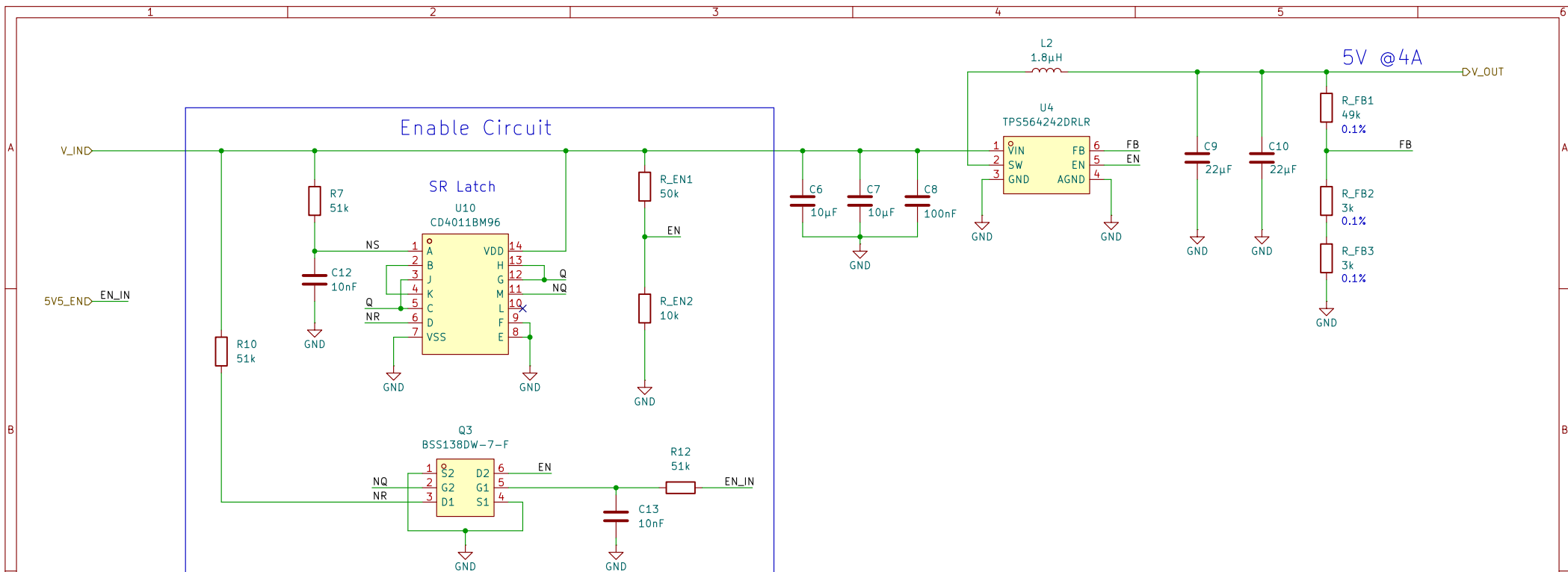
Title: POWER

Size: A4 Date: 2025-12-29

KiCad E.D.A. 9.0.6

Rev: V0.0

Id: 4/10



Notes

--- Output capacitance ---

$$C_{OUT} = 44\mu F \text{ (} 22\mu F, 100\mu F \text{) [typ., (min, max)]}$$

$$I_{L_CAP} = (V_o * (V_{in} - V_o)) / (\sqrt{12} * V_{in} * L * f_{sw}) = 0.39A$$

L -> Inductor value
fsw -> Switching frequency (1200kHz)

--- Inductance ---

$$L (V_o = 5V) = 1.8 \text{ (} 1.5, 4.7 \text{) } \mu H \text{ [typ., (min, max)]}$$

Inductance Ratings:

$$I_{p-p} = (V_o / V_{in_max}) * (V_{in_max} - V_o) / (L * f_{sw}) = 1.35A$$

$$I_{peak} = I_o + I_{p-p} / 2 = 5.35A$$

$$I_{L_RMS} = \sqrt{I_o^2 + (1/12) I_{p-p}^2} = 4.3A$$

--- Feedback Resistors ---

$$V_o = 0.6 * (1 + R_{FB1} / R_{FB2})$$

$$R_{FB2} = (0.6 + 0.6 R_{FB1}) / V_o$$

--- EN input ---

When:

- EN is > 1.18(1.07, 1.33)V the module is enabled.
- EN is < 1(0.95, 1.2)V the module is disabled.

$$\text{Fixing } R_{EN1} + R_{EN2} \rightarrow R_{EN2} = (R_{EN1} + R_{EN2}) * V_{EN} / V_{IN}$$

$$V_{CUT} = V_{EN} * (R_{EN1} + R_{EN2}) / R_{EN2}$$

To limit to 6V input with $R_{EN1} + R_{EN2} = 50k$:

$$R_{EN2} \sim 8.2k$$

$$R_{EN1} \sim 40k$$

$$V_{CUT} = 5.87V \text{ (} 5.58, 7.05 \text{)}$$

This is too unreliable so it will be driven by the MCU!!

BeepyRC

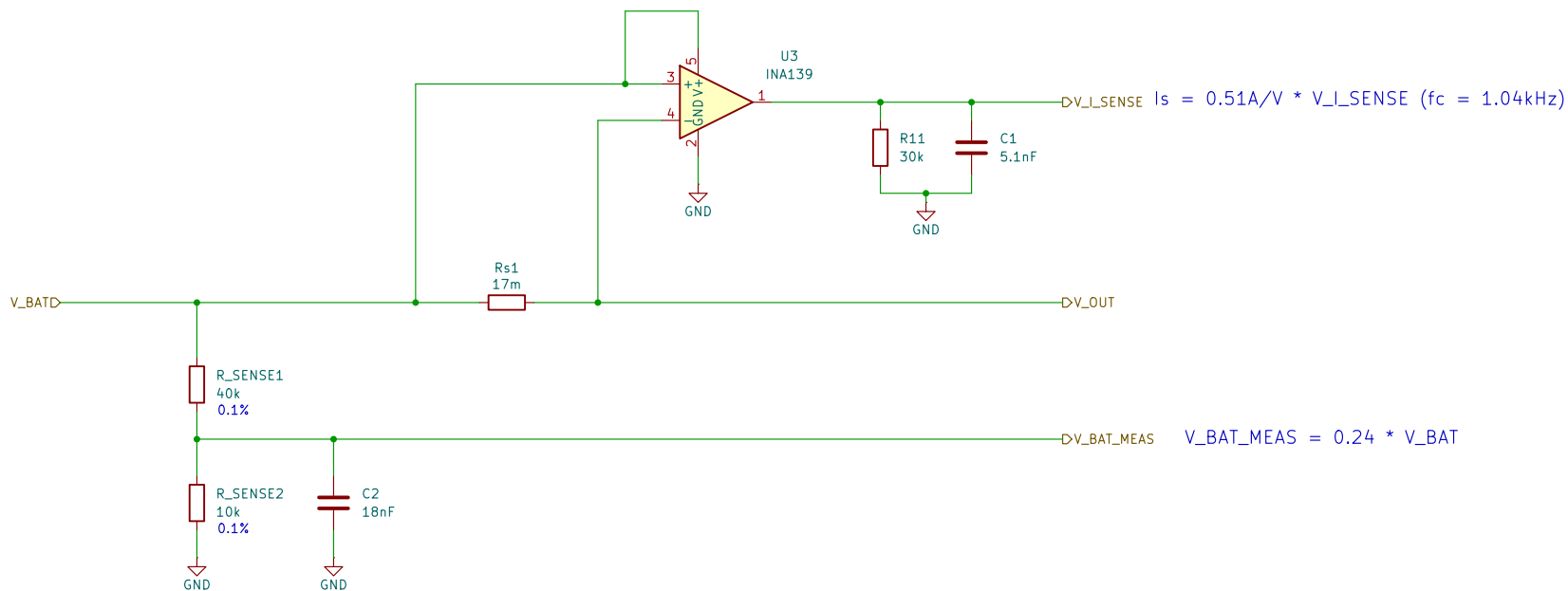
Electron Plant

Sheet: /Power/5VBuck/
File: buck_converter.kicad_sch

Title: Buck Converter

Size: A4 Date: 2026-02-08
KiCad E.D.A. 9.0.6

Rev: V0.0
Id: 5/10



--- Battery Voltage ---

2S battery range 6 – 8.4V, with 7.4 nominal
Maximum ranges 6 – 12V --> Limited by the N20 motors

R_SENSE is set to:

1. Be <50kOhm (ADC external resistance limit),
2. V_BAT_MEAS is ~3V @ 12V
3. Low-pass filter with Fc ~1kHz

--- INA139 ---

Output Voltage: $V_o = I_s * 1mA/V * R_L$

Vs full scale = 100mV (could also be 50mV to 500mV)

R_L -> Set to generate ~3V at 6A (<50kOhm, ADC external resistance limit)

C_L -> Low pass filter with a cut-off frequency at ~1kHz

--- Shunt Resistor ---

Vs = 100mV with max current

$\max(I_s) = 6A > (4 * 0.35A \text{ for the motors} + 4A \text{ for the 5.5V rail})$

$R_s = V_s / \max(I_s) = \sim 17m\Omega$

Power Rating: 0.6W

BeepyRC

Electron Plant

Sheet: /Power/BatteryMonitor/

File: bat_mon.kicad_sch

Title: Battery Monitor

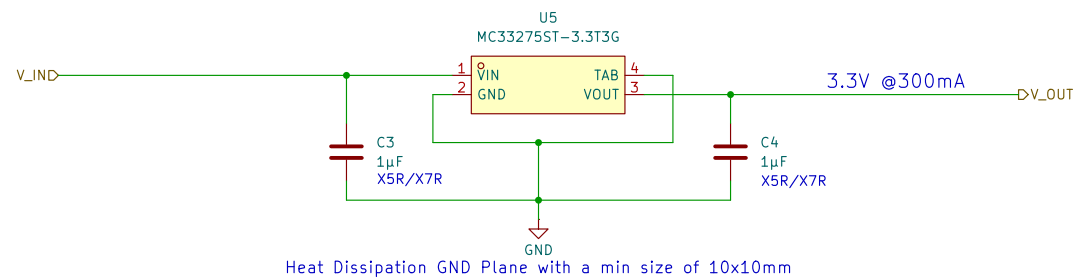
Size: A4

Date: 2026-02-11

Rev: V0.0

KiCad E.D.A. 9.0.6

Id: 6/10



BeepyRC

ElectronPlant

Sheet: /Power/3V3Regulator/
File: 3v_reg.kicad_sch

Title: 3V3 Regulator

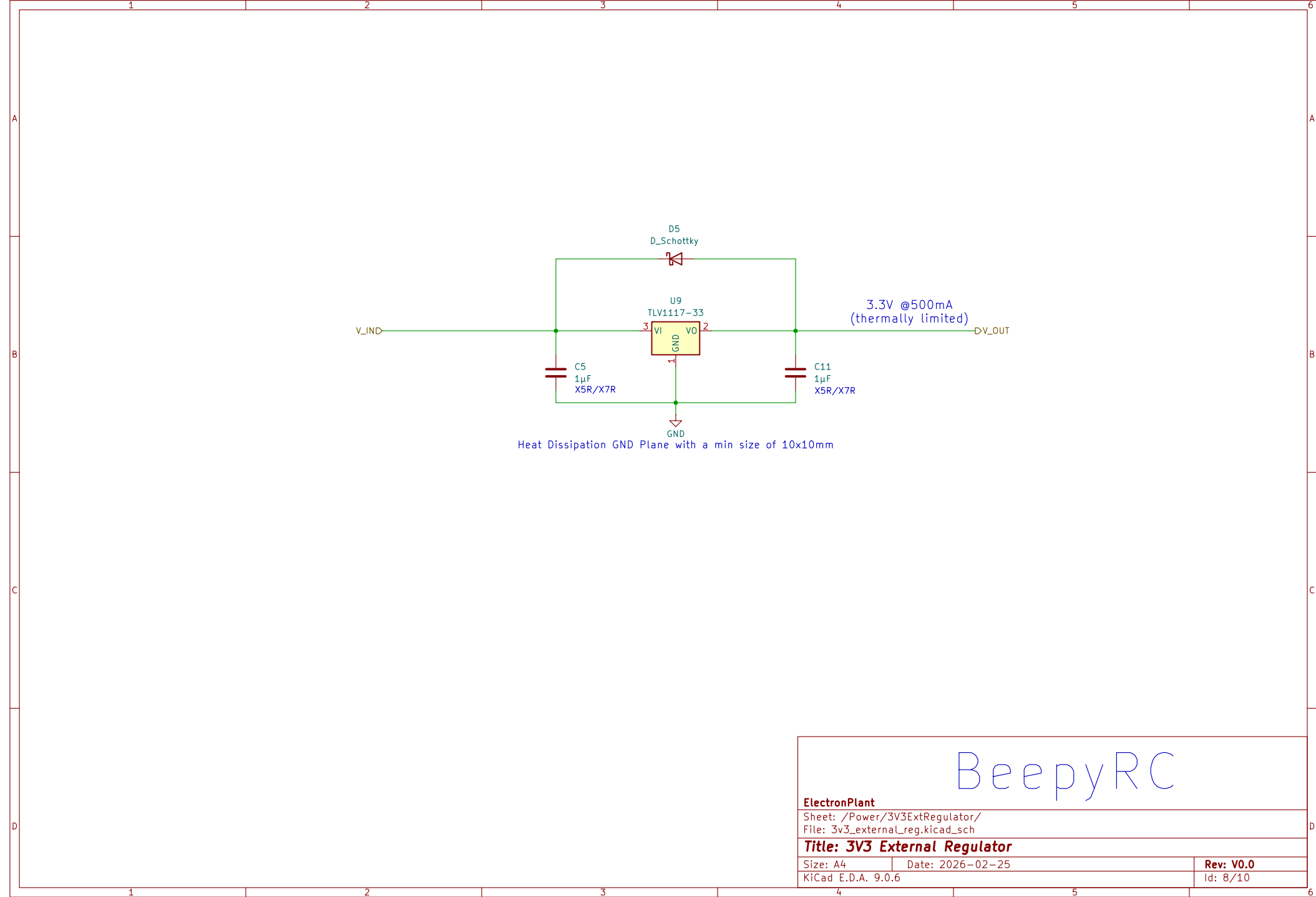
Size: A4

Date: 2026-02-25

Rev: V0.0

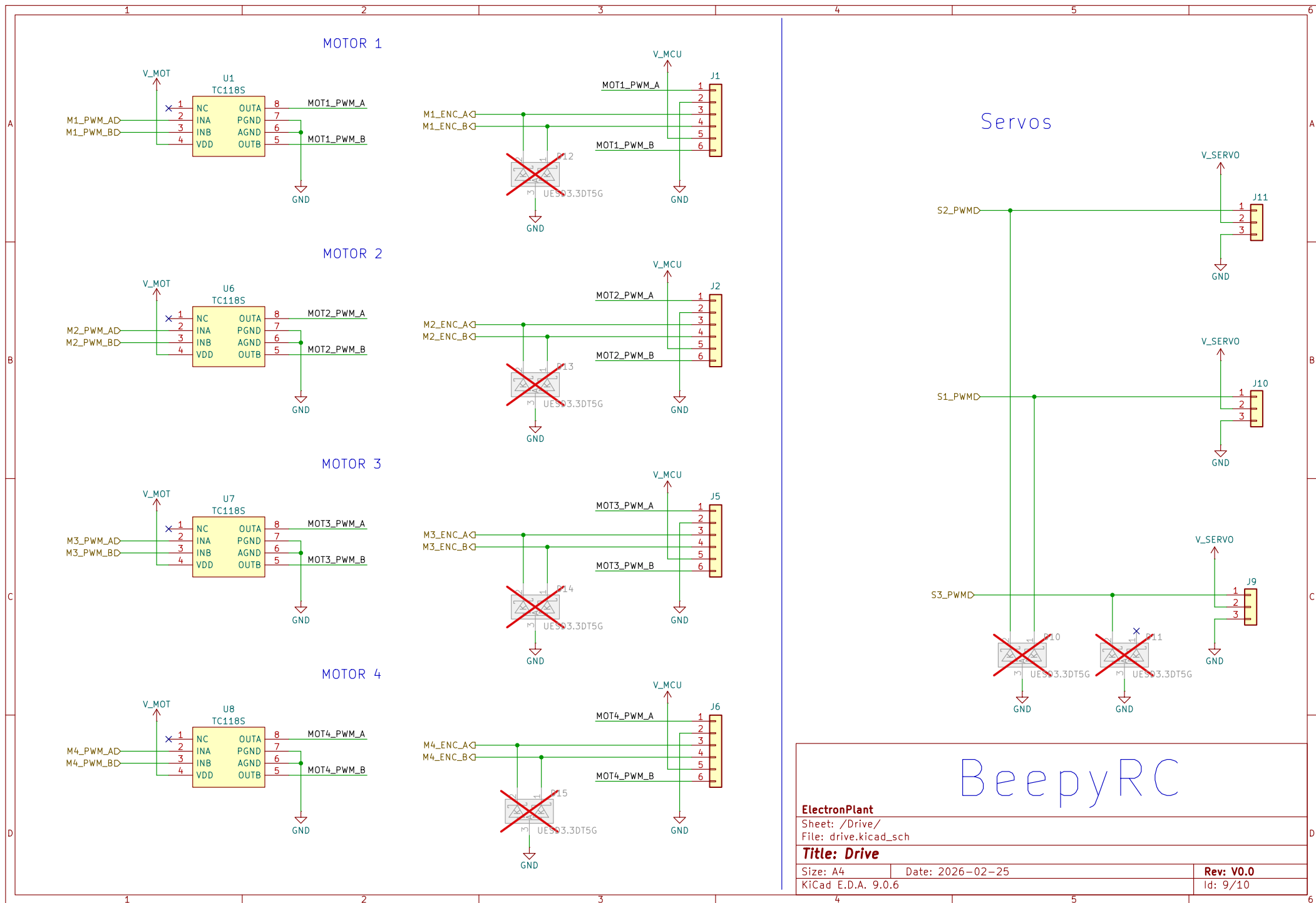
KiCad E.D.A. 9.0.6

Id: 8/10



BeepyRC

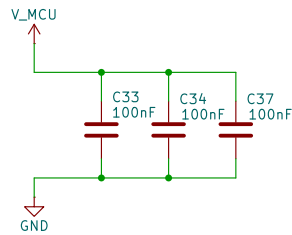
ElectronPlant		
Sheet: /Power/3V3ExtRegulator/		
File: 3v3_external_reg.kicad_sch		
Title: 3V3 External Regulator		
Size: A4	Date: 2026-02-25	Rev: V0.0
KiCad E.D.A. 9.0.6		Id: 8/10



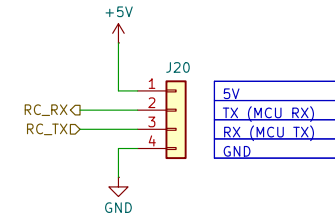
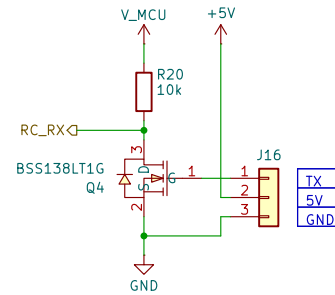
OSD

Decoupling strategy

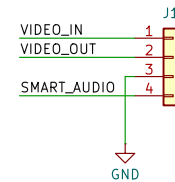
C14 between AVDD (P12) and AGND (P10)
C29 between DVDD (P2) and DGND (P1)
C30 between PVDD (P14) and PGND (P11)



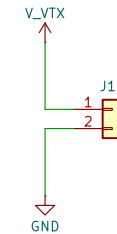
RC Serial



VTX

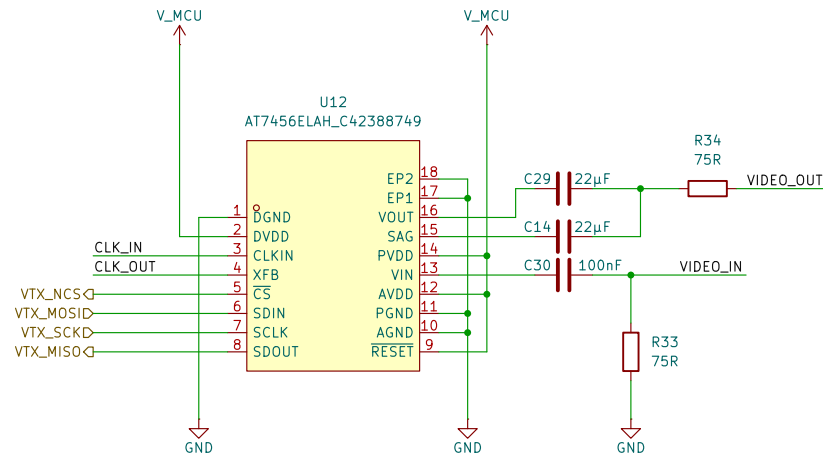


VTX Connector



Power Connector

CLK



BeepyRC

ElectronPlant

Sheet: /RC/
File: rc.kicad_sch

Title: RC

Size: A4
KiCad E.D.A. 9.0.6

Date: 2026-02-25

Rev: V0.0

Id: 10/10